Timothy Bielefeld

Herndon, VA | 703-638-9938 | timbeely@gmail.com | Portfolio
Embedded Control Systems & Firmware Development | Analog Circuit Design

Education

Purdue University – Bachelor of Science in Electrical Engineering – West Lafayette, IN – May 2028

- GPA: 3.88; Concentration in Automatic Control; Dean's List/Semester Honors, 2024-2026

Work Experience

Sports Concepts Lab – Electrical Engineering Intern – Reston, VA – May 2026 – August 2026

- **Wearable Haptic Feedback Device for Skill Reinforcement**
- **Hardware Design & Power:** Designed custom PCB schematics in KiCad powered by an ESP32-C3 microcontroller, incorporating rechargeable lithium-battery protection and power management to sustain >4 hours of active runtime.
- **Firmware & Control:** Wrote non-blocking C++ firmware that drives a 185 Hz Linear Resonant Actuator (LRA) via an I^2C haptic driver, utilizing auto-calibration routines to optimize motor feedback strength (2 G peak).
- **BLE & Mobile Application:** Developed a cross-platform mobile app using React Native, Expo, and TypeScript that streams multi-variable CSV instruction packets over Bluetooth Low Energy (BLE) to manage configurations across multiple athlete wearables simultaneously.

Projects

Optical Heart Rate Sensor, 2026

- Developed a non-invasive medical sensor using infrared emitters, phototransistors, and operational amplifiers to detect blood volume changes in capillaries.
- Formulated theoretical cutoff frequencies and gain equations, validating performance through Frequency Response Analysis (FRA), oscilloscope plots, and a physical LED indicator.

Smart HVAC System, 2026

- Built a closed-loop PID thermal regulation system utilizing an Arduino, motor driver, DC ventilation fan, and a temperature sensor. Implemented discrete PID loop control in C++ to compensate for thermal dead time and minimize overshoot, stabilizing target temperatures within ±0.5 °C.
- Included a local user interface using an LCD to allow real-time setpoint adjustment and live system state diagnostics.

Audio Equalizer, 2025

- Designed, simulated, and built a 3-band audio equalizer using op-amp gain stages and RC filters.
- Validated hardware cutoff frequencies with an oscilloscope, maintaining response within ±2 dB from 100 Hz–10 kHz.

Final Exam Calculator, 2026

- Created a responsive client-side web app (HTML/CSS/JS) hosted on GitHub Pages, utilizing dynamic Document Object Model (DOM) manipulation and LocalStorage API to calculate weighted grade thresholds and sustain user inputs across browser sessions.

Technical Skills

- **Embedded & Hardware:** Microcontrollers (ESP32, Arduino), PCB Design (KiCad), I^2C, BLE, PWM, Analog Circuit Design
- **Controls & Systems:** PID Loop Tuning, Feedback Control, Frequency Response Analysis (FRA)
- **Programming:** C/C++, Python (NumPy, SciPy), MATLAB, Java, TypeScript, HTML/CSS, Verilog
- **Tools:** Oscilloscope, LTSpice, Git/GitHub, Expo, React Native, SQL, Excel